

Machine-Learning Descriptor Analysis and Bayesian Data Augmentation for MXene Supercapacitor Capacitance Prediction: Implications for Physics-Guided Modelling

August 16, 2026

Abstract

Literature-derived datasets are an appealing route to data-driven design of MXene supercapacitor electrodes, but they are small and heterogeneous. We compile 118 usable gravimetric-capacitance observations for $\text{Ti}_3\text{C}_2\text{T}_x$ in H_2SO_4 from 40 publications, standardise thirteen structural and processing descriptors with unit-aware parsing, and evaluate prediction under strict publication-grouped cross-validation with a full target-leakage audit. Cross-paper transferability is weak ($R^2 = 0.03 \pm 0.12$, $\text{MAE} = 82 \text{ F g}^{-1}$). A variance decomposition localises the cause: 64% of the variance in capacitance and a mean of 87% of the variance in the descriptors lies *between* publications, and several descriptors are constant within each publication ($\text{ICC}(1) \approx 1$). We then test whether Bayesian-network augmentation (DATA SYNTHESIZER, correlated-attribute mode) relieves the small-sample constraint, fitting the generator inside each training fold only and sweeping four augmentation ratios, three network degrees and three generator seeds. It does not: the best condition improves MAE by 6.0 F g^{-1} , less than the 12.3 F g^{-1} obtained by re-drawing the synthetic set alone, with a non-monotonic response in the amount of synthetic data. The generator itself was faithful — no marginal rejected by a Kolmogorov–Smirnov test, Jensen–Shannon distance 0.08 on categorical frequencies, dependency structure reproduced to ≈ 0.06 , and no memorisation of real rows — so the null result reflects the nature of the bottleneck rather than a generator failure: resampling a learned joint distribution cannot supply information about experimental domains the source papers never visited. SHAP attribution is nevertheless highly stable under this resampling (Spearman $\rho = 0.90$, top-5 overlap 5/5), which licenses using the real-only ranking for descriptor prioritisation. Finally we assess what physics the corpus can enforce. A classical PDE-based physics-informed neural network is not justified — potential window, discharge time, current and any time axis are entirely absent — but two algebraic capacitance closure identities hold to a median relative error below 1%, allowing areal and volumetric capacitance to serve as auxiliary supervised outputs under a hard consistency constraint. We therefore recommend a hybrid physics-constrained multi-task regression, and identify the potential window and discharge time as the highest-value missing measurements.

Keywords: MXene; supercapacitor; gravimetric capacitance; SHAP; Bayesian network; synthetic data augmentation; domain shift; physics-guided machine learning.

1 Introduction

Two-dimensional $\text{Ti}_3\text{C}_2\text{T}_x$ MXene is among the most studied electrode materials for aqueous supercapacitors [1, 2], and a large body of work reports gravimetric capacitance in H_2SO_4 electrolyte. The accumulated literature is an attractive target for data-driven analysis: if the reported capacitances could be related to the structural and processing descriptors that accompany them, the resulting model would both rank the descriptors worth optimising and supply a data-driven prior for a subsequent physics-informed model.

That programme faces three obstacles that this study addresses directly. First, a corpus assembled from independent publications is not an experimental design. Cell geometry, reference electrode, mass-normalisation convention, potential window and reporting rate differ from paper to paper, and several descriptors are reported at most once per publication, so the unit of independent information is the paper rather than the row. Second, such corpora are small: after discarding every entry whose capacitance is reported only as a range or an inequality, 118 usable observations remain, drawn from 40 publications. Third, the descriptors of greatest physical interest — interlayer spacing, surface area, mass loading — are exactly those most often omitted.

Synthetic data augmentation is a natural response to the second obstacle. A recent study of flow-capacitive deionisation expanded 32 experimental observations to 200 using the correlated-attribute (Bayesian-network) mode of DATASYNTHESIZER and reported improved model performance [3]. Whether that strategy transfers to a corpus whose limiting factor is heterogeneity rather than row count is an open and practically important question, and it is the question this work tests.

The contributions are as follows. (i) A leakage-controlled, paper-grouped evaluation of gravimetric capacitance prediction on the compiled corpus. (ii) A pre-registered test of Bayesian-network augmentation at four ratios, three network degrees and three independent generator seeds, in which the generator is fitted inside each training fold only. (iii) A fidelity audit establishing that the generator reproduces the training distribution it was shown, so that the augmentation result can be interpreted rather than dismissed as a generator failure. (iv) A variance decomposition that locates the bottleneck in between-study heterogeneity. (v) A SHAP descriptor analysis [4, 5] whose ranking is shown to be stable under Bayesian resampling, together with an explicit assessment — rejecting a classical PINN [6] and specifying the physics-constrained alternative the data do support.

2 Dataset and preprocessing

2.1 Corpus

The corpus (`h2so4_corpus_final.csv`) contains 192 extracted electrode records with 83 fields, obtained by structured extraction from published articles reporting $\text{Ti}_3\text{C}_2\text{T}_x$ performance in H_2SO_4 . The prediction target is gravimetric capacitance C_g in F g^{-1} . Table 1 summarises the corpus after cleaning and Fig. 1 shows its structure.

2.2 Target cleaning

A record is retained only when its capacitance is a single defensible number. Values carrying an approximation marker (~ 325 , `about 1.1`) or a mean $\pm$ standard deviation are retained and flagged. Ranges (300–350) and inequalities (`over 200`, `<300`) are *never* collapsed to a midpoint: 4 such rows were discarded, alongside 70 rows reporting no value. 118 rows from 40 publications survive. Two physically extreme values (7.26 and 1609.5 F g^{-1}) are retained and flagged rather than deleted; a sensitivity analysis excluding them from scoring is reported in `02_real_only_outlier_sensitivity.csv`.

2.3 Descriptor standardisation

Thirteen descriptors were standardised by unit-aware parsing functions rather than manual editing. Conversions are applied only where the source unit is unambiguous ($\text{nm} \rightarrow \text{\AA}$, $\text{mm} \rightarrow \mu\text{m}$, $\mu\text{g cm}^{-2} \rightarrow \text{mg cm}^{-2}$, $\text{Vs}^{-1} \rightarrow \text{mVs}^{-1}$). Incompatible bases are left missing: a current density reported as mA cm^{-2} or A cm^{-3} is not convertible to A g^{-1} without information the corpus does not contain, and a mass loading given in mg or $\text{wt.}\%$ carries no area basis. Electrolyte molarity is parsed from the electrolyte field only; the synthesis field also contains acid concentrations (e.g. wet-spinning in 98 wt.% H_2SO_4) that are not the test electrolyte. Free-text morphology, composition and synthesis strings are mapped to categorical families. `synthesis_family` collapses to a single level (LiF/HCl MILD etching [7]) among the surviving rows and is therefore dropped as uninformative, leaving 12 descriptors.

2.4 Paper grouping and leakage control

Each row is assigned a `paper_id` (DOI where available, else source filename; the two are one-to-one in this corpus). Of 83 raw columns, 14 are admissible as predictors. Excluded are volumetric and areal capacitance — both algebraic transforms of the target — together with extraction-confidence and provenance metadata, free-text extraction rationales that can quote the target value verbatim, publication identifiers, and the `role/is_primary` annotations, which are assigned with knowledge of the measured performance. No energy- or power-density columns exist in this corpus, so the usual $E = \frac{1}{2}CV^2$ leakage route is absent.

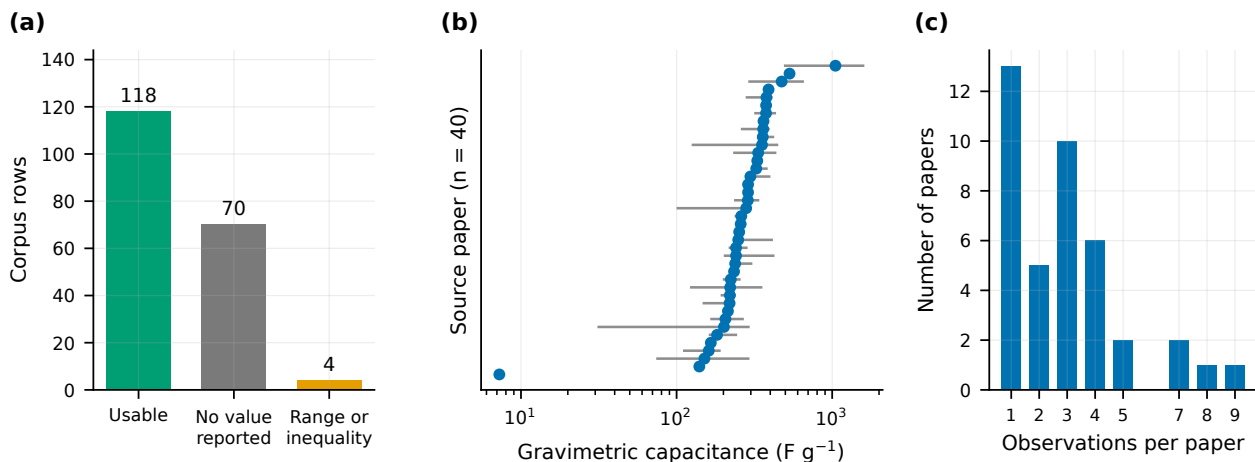


Figure 1: Structure of the compiled corpus. (a) Disposition of the 192 raw records. (b) Capacitance range of each of the 40 publications (bar: min-max; marker: median), sorted by median. (c) Distribution of observations per publication. The corpus is a set of small, tight within-paper clusters spread over a wide global range.

Table 1: Composition of the literature-derived H_2SO_4 MXene corpus after target cleaning and descriptor standardisation.

Quantity	Value
Rows in the raw corpus	192
Source publications in the raw corpus	49
Rows with a usable single-valued C_g	118
Publications represented after cleaning	40
Rows dropped: no value reported	70
Rows dropped: range or inequality literal	4
Approximate values retained and flagged	2
Median observations per paper	3
Maximum observations from one paper	9
Capacitance range (F g^{-1})	7.3–1609.5
Capacitance median (F g^{-1})	266.5
Standardised descriptors retained	12
Raw columns admissible as predictors	14

Range and inequality literals (e.g. 300–350, over 200) were discarded rather than converted to midpoints; every discarded row is logged in 01_target_disposition_log.csv.

3 Machine-learning methodology

3.1 Estimator

Gradient-boosted regression trees (XGBoost [8]) are used throughout. The estimator handles missing values natively, which matters here because several descriptors are reported by a minority of papers, and it is robust on the sample sizes involved. Categorical descriptors are one-hot encoded with an encoder fitted on the training partition only and `handle_unknown="ignore"`, so a level appearing solely in a held-out paper cannot influence the encoding.

3.2 Paper-grouped cross-validation

Ordinary random splitting is inadmissible here. Multiple rows originate from the same publication and share its cell, protocol and often its exact descriptor values, so a random split would place near-duplicates on both sides of the partition and report an optimistic score. All evaluation therefore uses `GroupKFold` on `paper_id` with five folds: rows from one publication are never split across training and test.

3.3 Hyper-parameter selection

Hyper-parameters are chosen by randomised search inside an *inner* paper-grouped split of the training partition of each outer fold, implemented with scikit-learn [9]. The outer test papers never influence model selection, encoding, or any preprocessing statistic. A fixed seed of 42 is used for NumPy, Python `random`, XGBoost, the cross-validation splitters and the synthetic generator.

3.4 Two models with different purposes

This is the central methodological distinction of the study and it is maintained throughout.

- The **paper-grouped cross-validation** above estimates *predictive transferability* to publications the model has not seen. It is the only number quoted as a performance estimate.
- A **final real-only interpretation model** is fitted to all 118 usable observations and is used *solely* to characterise multivariate descriptor associations within the compiled corpus. Its in-sample fit ($R^2 = 0.624$) is reported here only for completeness and is not a generalisation estimate.

Conflating the two would be the most consequential error available in this analysis, so the two models are named, stored and reported separately (`results_final/models/model_card.md`).

3.5 Metrics

R^2 , MAE and RMSE are reported per fold and aggregated as mean, SD and median. Because R^2 is normalised by the variance of each fold's own held-out targets, folds whose papers happen to agree with one another are scored against a very small denominator; pooled metrics computed over all held-out predictions jointly are therefore also reported.

4 Bayesian-network augmentation

4.1 Generator

Synthetic observations are produced with DATASYNTHESIZER 0.1.13 [10] in *correlated-attribute mode*, which fits a Bayesian network over the attributes by a greedy mutual-information search [11] and samples from the resulting factorisation. No substitute generator (CTGAN, Gaussian copula, SMOTE, KDE) was used at any point. The privacy budget is set to $\epsilon = 0$: the inputs are already-published literature values, so differential-privacy noise would degrade fidelity for no benefit.

4.2 Attribute treatment

The network is given 5 descriptors plus the target: `h2so4_M`, `composition_family`, `scan_rate_mV_s`, `interlayer_A` and `layer_class`. Electrolyte molarity and scan rate are declared *categorical* despite being numeric: each occurs at only a handful of discrete experimenter-chosen settings, and a continuous treatment would let the generator invent molarities and sweep rates that no experiment in the corpus used. Interlayer spacing and the target remain continuous.

4.3 Cohort

Requiring every selected descriptor to be reported in the same publication reduces the corpus to a complete-case cohort of 47 rows from 18 papers. **All four training conditions use exactly this cohort and this feature set**, so the comparison is like-for-like. This cohort model is a control arm for the augmentation experiment and is not the headline real-only baseline of Sec. 5.

4.4 Anti-leakage protocol

Within each outer grouped fold the Bayesian network is fitted on the *training papers only*; synthetic rows are drawn from that network; the physical-validity filter uses training-fold minima and maxima only; the one-hot encoder and the hyper-parameter search also see training papers only. Held-out papers are never described, never generated from, never augmented, and never enter any training set. No synthetic row ever appears in a test partition.

4.5 Experimental grid

Four training conditions are compared: real only, and real plus $1\times$, $3\times$ and $5.25\times$ synthetic rows, the last reproducing the $200/32 = 6.25$ total-size ratio of the motivating study. Network degree $k \in \{1, 2, 3\}$ is swept, with $k = 2$ as the pre-registered default. Each augmented condition is generated three times with independent generator seeds inside every fold and the fold score is the mean over those draws, so that re-drawing the synthetic set cannot masquerade as an effect. Generated rows pass a physical-validity filter (positive capacitance, non-negative descriptors, categorical levels seen in training, values inside the training-fold range) before use.

5 Cross-paper prediction results

Table 2 and Fig. 2 report the paper-grouped cross-validation of the real-only model on the full descriptor set. Averaged over folds, $R^2 = 0.030 \pm 0.119$, $\text{MAE} = 81.5 \pm 32.8 \text{ F g}^{-1}$ and $\text{RMSE} = 129.2 \pm 84.9 \text{ F g}^{-1}$; pooled over all held-out predictions, $R^2 = 0.081$. Fold-level R^2 ranges from -0.102 to 0.151.

Interpretation of a near-zero and partly negative R^2 . A negative R^2 on a fold means the predictions for those held-out papers are worse, under the coefficient-of-determination criterion, than simply predicting the mean of that fold’s own test targets — a baseline unavailable in practice, since it requires the answers. It does not mean the model is uninformative in absolute terms: the mean absolute error of 82 F g^{-1} against a corpus median of 266 F g^{-1} is poor but not vacuous. What the result does establish is that *cross-paper predictive transferability is weak*. We report this plainly; it is the central empirical constraint on everything that follows.

5.1 Why: cross-study heterogeneity

A one-way variance decomposition by publication (Table 3) locates the difficulty. For the target, the between-paper share of total sum-of-squares is $\eta^2 = 0.64$ with an intraclass correlation of $\text{ICC}(1) = 0.47$: most of the variation in capacitance is variation between publications, not within them.

The descriptors are far more paper-bound still, averaging $\eta^2 = 0.87$. Several are *paper-level constants*: electrolyte molarity, gravimetric current density and pore diameter have $\text{ICC}(1) \approx 1$, and layer morphology and synthesis route take a single level within essentially every publication (mean within-paper purity 1.00). The median within-paper range of a numeric descriptor is only 0% of its global range.

This has a direct consequence for the cross-validation. Holding out a publication removes an entire descriptor setting rather than a sample from a shared distribution, so the model is asked to extrapolate to a corner of descriptor space it has never observed. Per fold, the fraction of held-out capacitances lying inside the training range is 0.92–1.00, and the standardised mean difference between training and test targets reaches 0.12.

It also has a direct consequence for interpretation, developed in Sec. 8: when a descriptor is constant within each paper, its apparent importance is statistically inseparable from the identity of the papers that used that setting.

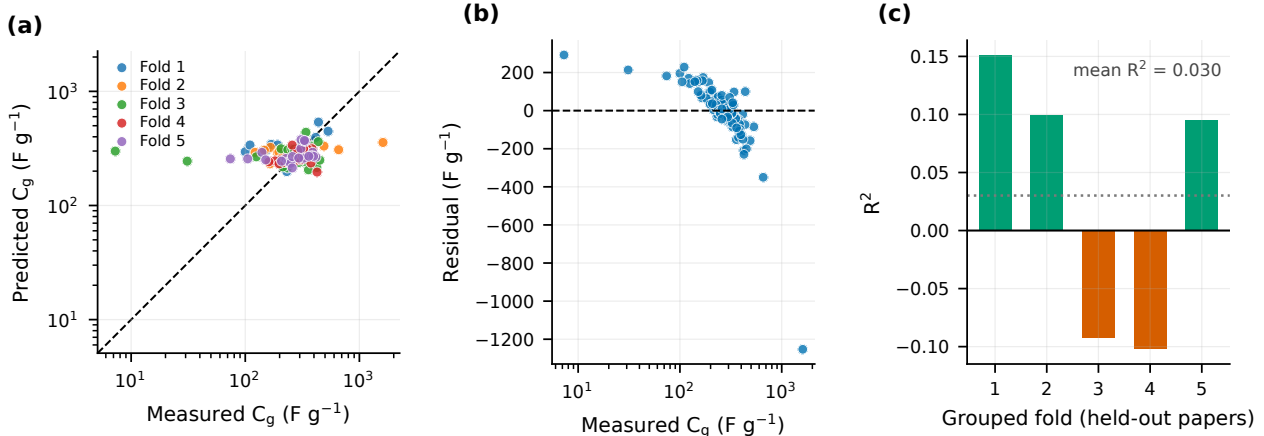


Figure 2: Paper-grouped cross-validation of the real-only model. (a) Measured versus predicted capacitance for held-out papers, coloured by fold; the dashed line is parity. (b) Residuals. (c) Per-fold R^2 ; the dotted line is the mean. Predictions regress towards the corpus centre, which is the expected behaviour when each fold requires extrapolation to an unobserved descriptor setting.

Table 2: Paper-grouped cross-validation of the real-only XGBoost model on the full descriptor set. MAE and RMSE in F g⁻¹. Rows from one publication never appear in both the training and test partition.

Fold	n_{train}	n_{test}	Papers	R^2	MAE	RMSE
1	94	24	8	0.151	61.3	90.4
2	94	24	8	0.099	134.7	278.7
3	94	24	8	-0.092	90.3	115.7
4	95	23	8	-0.102	52.4	73.9
5	95	23	8	0.095	69.0	87.5
Mean				0.030	81.5	129.2
SD				0.119	32.8	84.9
Median				0.095	69.0	90.4
Pooled				0.081	81.9	150.8

“Pooled” scores every held-out prediction from every fold together, which avoids normalising by a near-zero within-fold target variance while keeping the grouped hold-out intact.

6 Bayesian augmentation results

Table 4 and Fig. 3 compare the four training conditions on the identical set of held-out real papers.

Real-only training on the cohort gives $\text{MAE} = 129.6 \pm 68.0 \text{ F g}^{-1}$. The nominally best augmented condition (D, +5.25 $\times$) reaches 123.6 F g^{-1} , an improvement of 6.0 F g^{-1} (-4.6%), improving MAE in 3 of 5 folds. The +3 $\times$ condition is the *worst* of the four.

6.1 Why this improvement is not robust

Three observations, all visible in Fig. 4, place the apparent gain inside the configuration noise.

1. **Re-drawing the synthetic set moves the score by more than the effect.** Holding fold, ratio and network degree fixed and changing only the generator seed shifts fold MAE by 12.3 F g^{-1} on average — larger than the 6.0 F g^{-1} attributed to the best ratio.

Table 3: One-way variance decomposition by source publication. η^2 is the share of total sum-of-squares lying between papers; ICC(1) is the corresponding intraclass correlation corrected for unequal group sizes.

Variable	Role	n	Papers	η^2	ICC(1)
current density A g	descriptor	35	16	1.000	1.000
pore diameter nm	descriptor	4	4	1.000	–
h2so4 M	descriptor	111	38	1.000	1.000
ssa m2 g	descriptor	35	17	0.989	0.980
interlayer A	descriptor	88	32	0.970	0.955
electrode thickness um	descriptor	54	20	0.969	0.955
flake size um	descriptor	11	6	0.727	0.493
scan rate mV s	descriptor	88	29	0.672	0.526
target cap F g	target	118	40	0.645	0.474
mass loading mg cm2	descriptor	55	19	0.500	0.263

Values approaching unity mean a variable is essentially constant within a publication and varies only between publications, so holding out a whole paper requires the model to extrapolate rather than interpolate.

- Changing the network degree moves it as much as changing the ratio.** At fixed ratio, varying $k \in \{1, 2, 3\}$ changes MAE by up to 17.8 F g^{-1} , while the entire spread across the four augmentation ratios is 15.1 F g^{-1} . Two nuisance parameters and the treatment produce effects of the same magnitude.
- The response is not monotone in the amount of synthetic data.** The ordering is $+5.25\times$, real-only, $+1\times$, $+3\times$. A genuine regularisation benefit would not place the middle ratio last.

Paired Wilcoxon signed-rank tests across the five folds were computed (`06_statistical_comparison.csv`), but with five paired folds the smallest attainable two-sided p -value is $2^{-4} = 0.0625$, so no comparison here can reach $p < 0.05$ by construction. The p -values are reported for completeness only; the effect direction, its consistency across folds and its size relative to the noise floor are the meaningful quantities, and none supports the augmentation.

Conclusion. Bayesian-network augmentation did not produce robust evidence of improved prediction on unseen real papers at any ratio tested. This is reported as found: no configuration search was run to make the synthetic data appear beneficial, and the largest augmentation ratio was not preferentially reported.

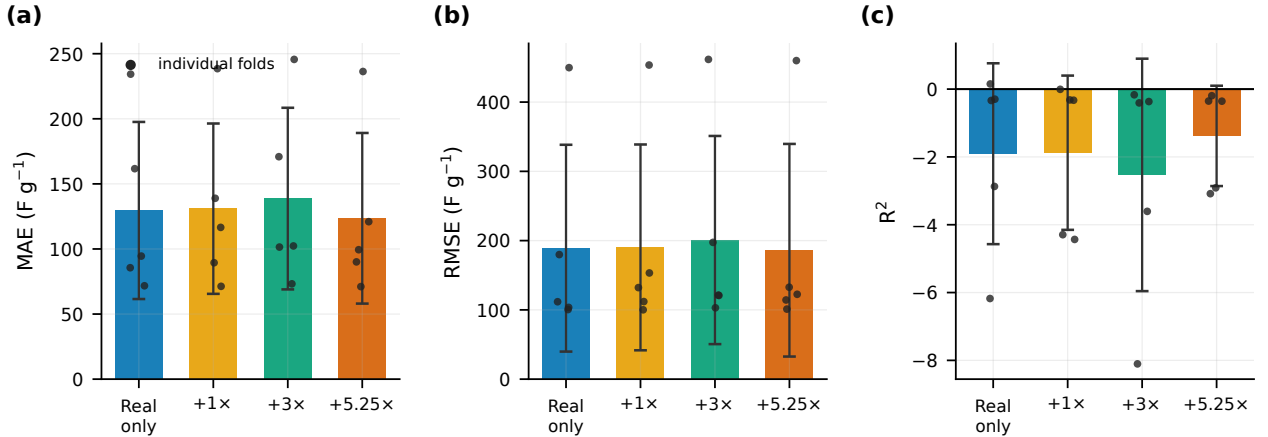


Figure 3: Real-only versus augmented training on unseen real papers ($k = 2$; bars are means over five grouped folds, error bars the fold-to-fold SD, points the individual folds). (a) MAE, (b) RMSE, (c) R^2 . The conditions are indistinguishable relative to the fold-to-fold spread.

7 Synthetic-data fidelity

A null augmentation result is only informative if the generator itself worked. Every metric below compares synthetic rows against the real training rows *of the same fold*, never the test papers (Table 5, Fig. 5).

Marginals. The mean Kolmogorov–Smirnov statistic is 0.154 and **no** continuous variable was rejected at $p < 0.05$ in any fold, ratio or seed.

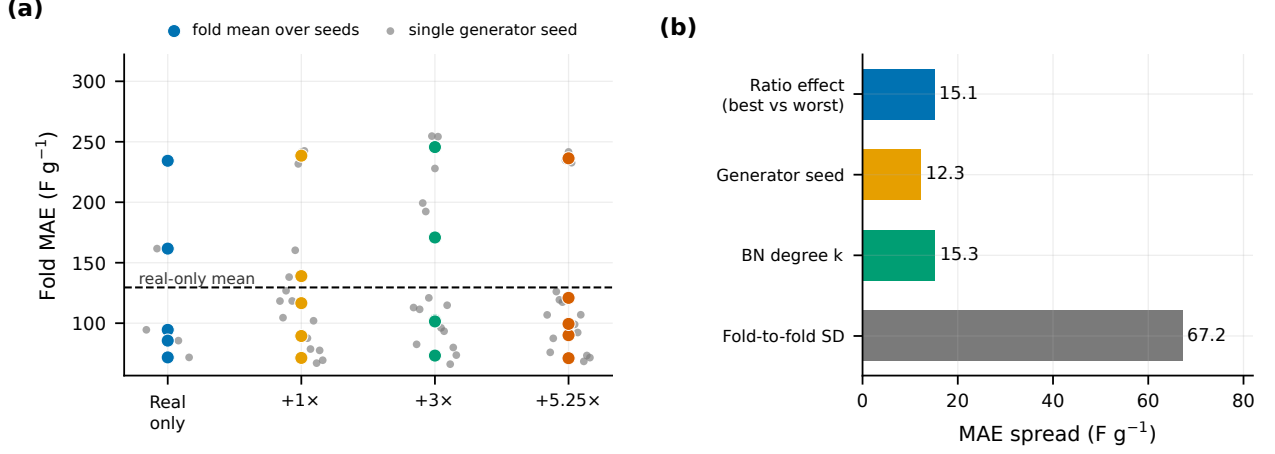


Figure 4: The apparent augmentation effect against the noise it must be judged against. (a) Fold MAE for every generator seed (grey) and the fold mean over seeds (coloured); the dashed line is real-only training. (b) The best-versus-worst ratio effect compared with the spread produced by re-drawing the synthetic set, by changing the Bayesian-network degree, and by fold-to-fold variation. The treatment effect is indistinguishable in magnitude from the two nuisance sources and is an order of magnitude below the fold-to-fold spread.

Table 4: Real-only versus Bayesian-augmented training, evaluated on the identical set of held-out real papers in every condition ($k = 2$, five grouped folds, three independent generator seeds averaged per fold). Errors in $F g^{-1}$; $\pm$ denotes the fold-to-fold SD.

Training set	n_{syn}	R^2	MAE	RMSE	Pooled MAE	Seed SD
Real only	0	-1.91 ± 2.67	129.6 ± 68.0	189.1 ± 149.3	129.5	–
Real + 1x	38	-1.88 ± 2.27	130.9 ± 65.4	190.2 ± 148.6	130.8	11.1
Real + 3x	113	-2.53 ± 3.43	138.7 ± 69.8	200.8 ± 150.3	138.6	18.9
Real + 5.25x	198	-1.38 ± 1.48	123.6 ± 65.5	186.1 ± 153.5	122.8	6.9

“Seed SD” is the within-fold MAE standard deviation across the three generator seeds, i.e. the score movement produced by re-drawing the synthetic set alone. It is of the same magnitude as the entire spread between augmentation ratios, and for the +3x condition it exceeds it.

Categorical frequencies. The mean Jensen–Shannon distance is 0.079, indicating close agreement of level frequencies.

Dependency structure. The Pearson correlation matrix differs by 0.047 in mean absolute value and the normalised mutual-information matrix, which also covers the categorical descriptors, by 0.055.

Descriptor–target relationships. This is the fidelity criterion that matters most for downstream utility, since a generator can match every marginal and still destroy the relationship a regressor must learn. Measured as Spearman correlation with the target for continuous descriptors and η^2 for categorical ones, the mean absolute change is 0.065.

Memorisation. Across all folds, ratios and seeds, 0 synthetic rows were exact copies of a real training row and 0 were duplicated within a synthetic set. The median synthetic-to-real nearest-neighbour distance is 0.33 times the median real-to-real distance. A ratio below unity is expected rather than alarming here, since 37–200 synthetic points are drawn over the same manifold as 37 real ones and nearest-neighbour distances contract with density alone; the sharper diagnostic is that only 3.5 of 116 synthetic rows on average fall closer to a real record than the fifth percentile of real-to-real spacing.

Central interpretation. The generator reproduced the joint distribution it was shown, and did so without copying. **Distributional fidelity to the training fold does not imply improved generalisation to a held-out publication.** A Bayesian network resamples the dependencies present in its training data; it cannot supply information about experimental domains the training papers never visited. Since Sec. 5.1 establishes that the held-out papers are precisely such unvisited domains, the null result of Sec. 6 is the expected outcome rather than an anomaly, and it is not attributable to a defective generator.

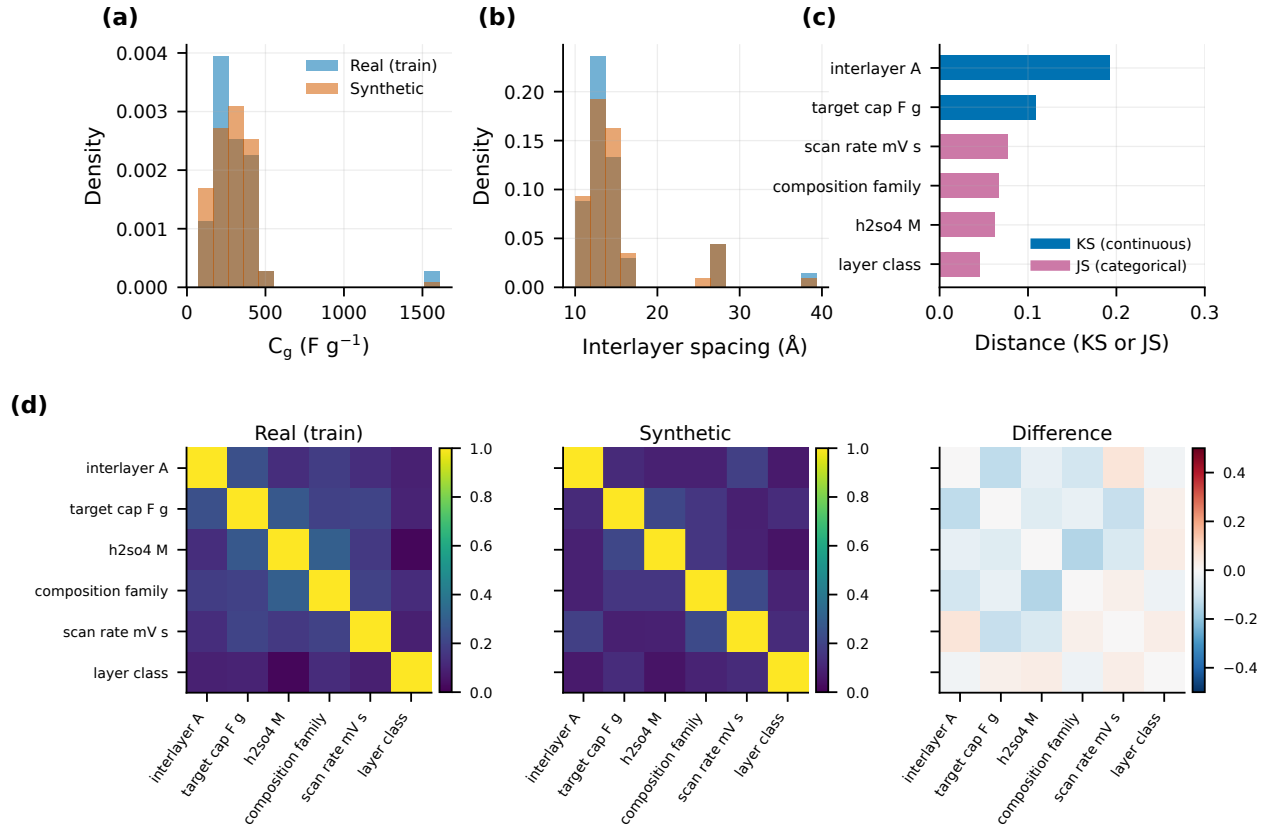


Figure 5: Fidelity of the Bayesian-resampled observations against the real training rows of the same fold. (a) Target marginal, (b) interlayer-spacing marginal, (c) per-variable Kolmogorov–Smirnov (continuous) and Jensen–Shannon (categorical) distances, (d) normalised mutual-information matrices for real and synthetic data and their difference.

Table 5: Fidelity of the Bayesian-network synthetic observations, computed against the real training rows of the same fold ($k = 2$; mean, min and max over folds, augmentation ratios and generator seeds).

Fidelity metric	Mean	Min	Max
Mean KS statistic (continuous)	0.154	0.095	0.198
Continuous variables with KS $p < 0.05$	0.0	0.0	0.0
Mean Jensen–Shannon distance (categorical)	0.079	0.028	0.170
Pearson correlation matrix, mean $ \Delta $	0.047	0.002	0.346
Spearman correlation matrix, mean $ \Delta $	0.095	0.001	0.292
Mutual-information matrix, mean $ \Delta $	0.055	0.025	0.093
Descriptor–target association, mean $ \Delta $	0.065	0.010	0.263
Exact duplicates of a real training row	0.0	0.0	0.0
Duplicates within a synthetic set	0.0	0.0	0.0
Synthetic→real NN distance (median)	0.0065	0.0046	0.0086
Real→real NN distance (median)	0.0195	0.0130	0.0351

Nearest-neighbour distances use a range-normalised mixed-type (Gower) metric. No synthetic row reproduced a real training row exactly.

8 Real-only SHAP descriptor analysis

SHAP values [4] are computed with `TreeExplainer` [5] on the final real-only interpretation model fitted to all 118 usable observations (Sec. 3). One-hot columns are reported individually and aggregated back to the parent descriptor by summing $|\text{SHAP}|$ within each row before averaging. This is an *association* analysis within the compiled corpus, not a causal or predictive claim.

8.1 Descriptor ranking

Table 6 and Fig. 6 give the ranking. The five leading descriptors, with their share of total attribution, are: composition family (22%); H_2SO_4 concentration (M) (20%); scan rate (mV s^{-1}) (16%); interlayer spacing ($\text{\AA}$) (10%); layer morphology (9%). The ranking agrees closely with the independent fold-wise held-out SHAP computed during cross-validation: the top five descriptors each entered the fold-level top five in 80–100% of folds.

8.2 Direction of the associations

Fig. 7 and Fig. 8 show directionality. Higher H_2SO_4 concentration is associated with higher predicted capacitance ($\rho = +0.72$ between the descriptor value and its own SHAP value), consistent with greater proton availability for Ti–O pseudocapacitance [12]. Higher scan rate is associated with lower predicted capacitance ($\rho = -0.93$), consistent with rate-limited ion access to the interlayer galleries.

A trend that contradicts the naive expectation. Interlayer spacing is associated with *lower* predicted capacitance ($\rho = -0.75$), the opposite of the expectation that a wider gallery eases ion access. We do not report this as a physical finding. Two confounds are more plausible: the largest reported d -spacings in this corpus belong predominantly to pillared and composite architectures whose *gravimetric* capacitance is diluted by the mass of the intercalant, and d -spacing is conventionally measured on a dry film rather than in the hydrated, polarised state in which charge is actually stored. This is a concrete illustration of why a monotonicity prior must not be read off a SHAP plot.

The confounding caveat, stated quantitatively. Sec. 5.1 showed that several descriptors are paper-level constants. Electrolyte molarity, the second-ranked descriptor, has $\text{ICC}(1) = 1.00$: it does not vary at all within a publication. Its SHAP attribution therefore cannot be statistically separated from the identity of the publications that used each molarity, and the same applies to interlayer spacing ($\text{\AA}$), current density (A g^{-1}), electrode thickness (μm) and specific surface area ($\text{m}^2 \text{g}^{-1}$). The ranking should be read as “descriptors along which the corpus is organised and which the model exploits”, not as “levers that would change capacitance if turned”. Distinguishing the two requires controlled experiments that vary one descriptor within a single laboratory protocol — which is precisely what the corpus lacks.

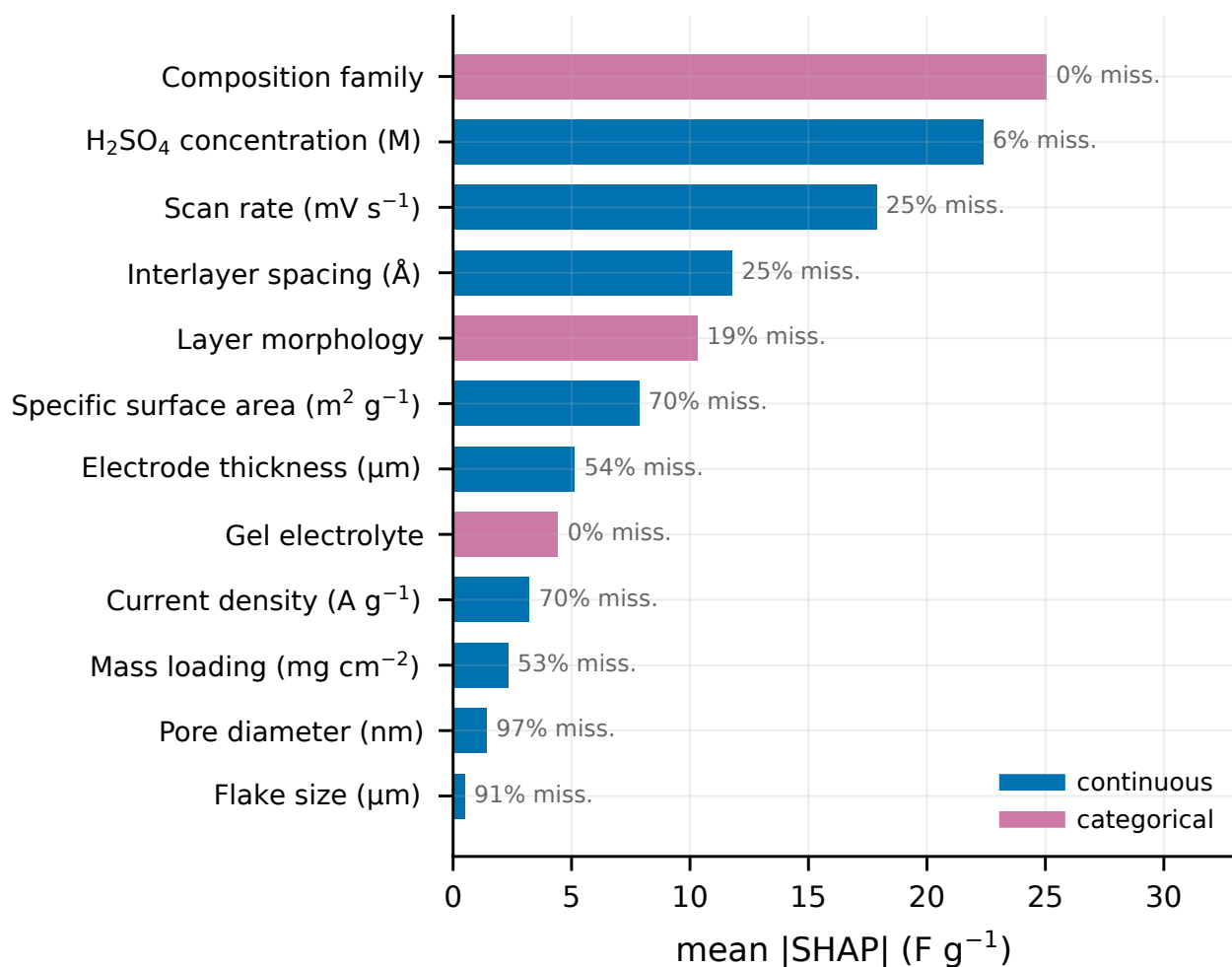


Figure 6: Descriptor-level SHAP importance of the final real-only interpretation model. Bars are the mean absolute SHAP value; annotations give the fraction of observations for which the descriptor is not reported.

Table 6: Descriptor-level SHAP attribution of the final real-only interpretation model fitted to all usable real observations. |SHAP| in F g⁻¹; share, missingness and top-5 frequency in per cent.

Rank	Descriptor	SHAP	Share	Miss.	Fold rank	Rank SD	Top-5	ρ
1	Composition family	25.05	22.3	0	2.4	0.89	100	–
2	H ₂ SO ₄ conc. (M)	22.37	19.9	6	1.4	0.89	100	0.72
3	Scan rate (mV s ⁻¹)	17.87	15.9	25	2.8	1.10	100	-0.93
4	Interlayer spacing (Å)	11.78	10.5	25	4.0	1.58	80	-0.75
5	Layer morphology	10.31	9.2	19	4.8	0.45	100	–
6	Specific surface area (m ² g ⁻¹)	7.85	7.0	70	6.4	1.52	20	0.69
7	Electrode thickness (μm)	5.14	4.6	54	7.0	1.00	0	-0.23
8	Gel electrolyte	4.43	3.9	0	10.0	2.00	0	–
9	Current density (A g ⁻¹)	3.23	2.9	70	7.8	1.48	0	0.04
10	Mass loading (mg cm ⁻²)	2.33	2.1	53	9.6	0.89	0	-0.07

“Fold rank” and “Rank SD” come from the independent held-out SHAP computation of the paper-grouped cross-validation; “Top-5” is the fraction of folds in which the descriptor entered the five most important. ρ is the Spearman correlation between a descriptor’s value and its own SHAP value (direction of the association, not a causal claim); it is defined for continuous descriptors only.

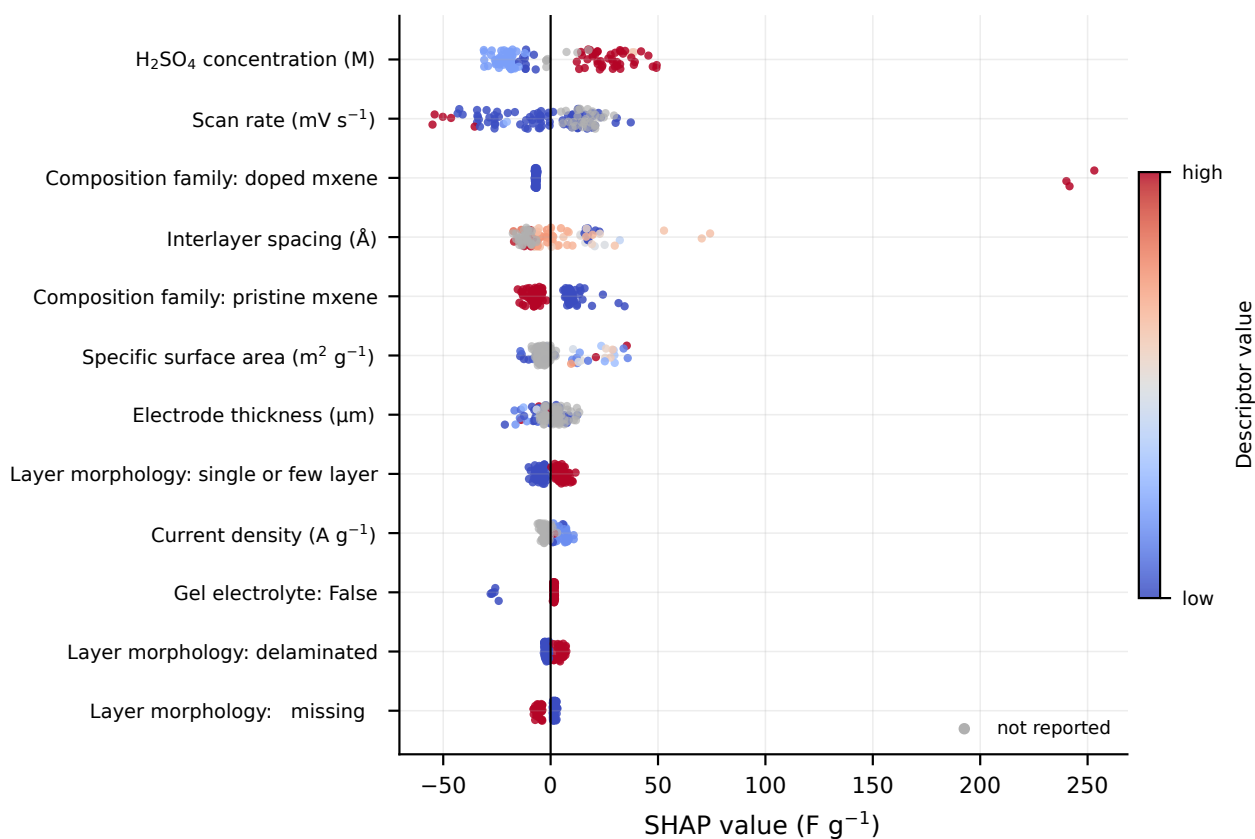


Figure 7: SHAP value distribution per encoded feature, coloured by descriptor value. Grey points are observations for which the descriptor is not reported.

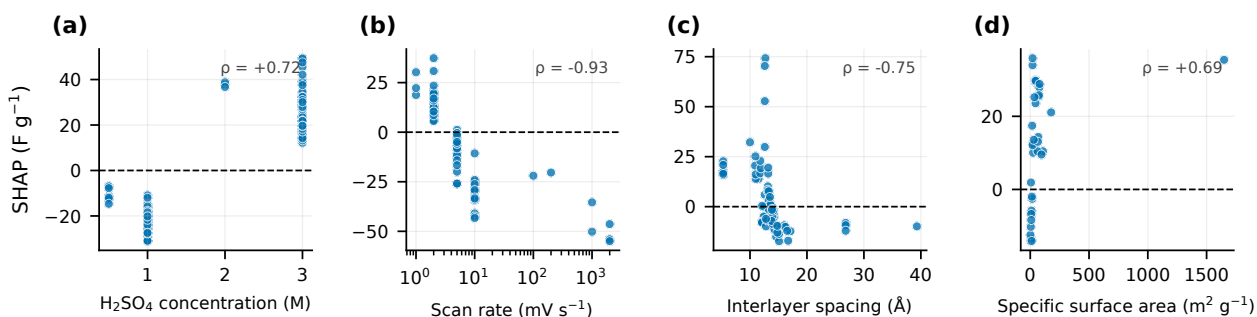


Figure 8: SHAP dependence for the leading continuous descriptors. ρ is the Spearman correlation between the descriptor value and its own SHAP contribution. The negative trend in interlayer spacing runs against the naive physical expectation and is discussed in the text.

9 SHAP robustness under augmentation

Although augmentation did not improve prediction, it provides a useful perturbation test: if the descriptor hierarchy survives having the training set resampled from a learned Bayesian network, it is not an artefact of the particular rows sampled.

Comparing the cohort real-only model with the best-performing augmented model on the identical cohort and feature set (Fig. 9):

- Spearman rank correlation of descriptor rankings: $\rho = 0.90$ ($p = 0.037$).
- Top-3 overlap 3/3; top-5 overlap 5/5.
- Mean absolute rank displacement 0.4 positions (maximum 1).
- Mean absolute change in each descriptor’s share of total attribution: 0.018.

The descriptor hierarchy is therefore stable. This matters for the argument of this paper: the same experiment that fails to improve prediction succeeds in demonstrating that the *relative* descriptor ordering is robust to resampling of the training distribution. Had augmentation improved R^2 while substantially reordering the descriptors, that would have indicated synthetic-distribution distortion and the ranking would not be usable; the observed combination — no predictive gain, high attribution stability — is the one that licenses using the real-only SHAP ranking for descriptor selection.

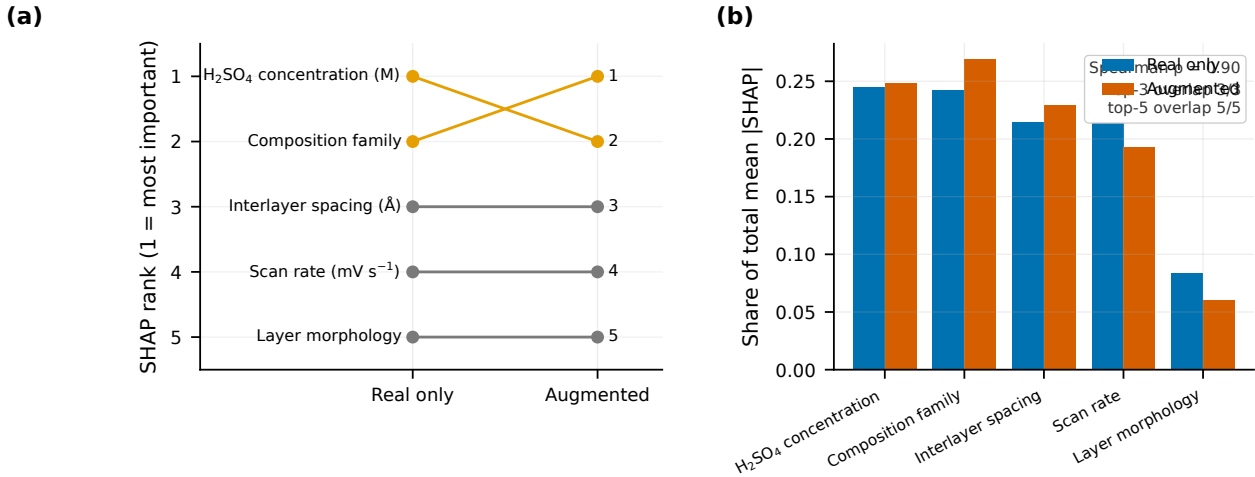


Figure 9: Descriptor attribution before and after Bayesian augmentation. (a) Rank slope chart; grey lines indicate unchanged rank. (b) Share of total mean |SHAP| per descriptor.

10 Implications for physics-informed modelling

10.1 Which quantities the corpus actually contains

Table 7 audits the corpus against the quantities required by the standard capacitance relations. The galvanostatic definition

$$C_g = \frac{I \Delta t}{m \Delta V} \quad \text{equivalently} \quad C_g = \frac{j \Delta t}{\Delta V}, \quad (1)$$

and the cyclic-voltammetry relation

$$C_g = \frac{\int I(V) dV}{m \nu \Delta V}, \quad (2)$$

both require a potential window ΔV . That quantity, together with the discharge time Δt , the absolute current I , the absolute electrode mass m , the electrode area and the integrated CV current, is **entirely absent** from the corpus (`electrode_area_cm2`, `voltage_window_V`, `current_A`, `discharge_time_s`, all at 0% coverage). The corpus does not even record whether a given row is a galvanostatic or a voltammetric measurement.

10.2 Assessment of three candidate model classes

(A) Classical PDE/ODE-based PINN — not justified. A physics-informed neural network in the usual sense [6] minimises the residual of a differential equation at collocation points, which requires a state variable defined over a space or time coordinate together with initial and boundary conditions per observation. Every row in this corpus is a single scalar summary of a completed experiment: there is no time axis, no potential axis, no spatial coordinate and no trajectory. Without ΔV and Δt even the *algebraic* relation (1) cannot be evaluated for a single row, let alone a differential form. Posing a PDE residual here would mean inventing physics rather than enforcing it, and we therefore reject this option explicitly.

(B) Physics-guided neural network — feasible. Positivity of capacitance, training-domain bounds and dimensionally meaningful inputs are all available. Monotonicity constraints, however, should *not* be imposed: the corpus shows an interlayer-spacing trend opposite to the naive expectation (Sec. 8), and imposing a monotone prior from SHAP would encode a probable confound as physics.

(C) Hybrid physics-constrained multi-task regression — recommended. Two exact algebraic closure identities relate the target to other measured quantities:

$$C_A = C_g m_A, \quad (3)$$

$$C_V = C_g \rho, \quad \rho = m_A/t, \quad (4)$$

where m_A is areal mass loading, t electrode thickness and ρ the electrode density. We verified both against the values reported in the source publications (Table 8). Equation (3) reproduces the reported areal capacitance with a median relative error of 0.38% and holds within 5% for 85% of 39 testable rows; Eq. (4) has a median error of 0.87% and holds within 5% for 76% of 34 rows.

This yields a structural insight that reframes two of the excluded columns. Areal and volumetric capacitance cannot be *predictors* — they are algebraic transforms of the target and were correctly removed as leakage — but they can serve as **auxiliary supervised outputs**, tied to the primary output by Eqs. (3) and (4). Two leakage columns thereby become additional training signal and a genuine internal-consistency constraint, with no leakage introduced.

10.3 Descriptor classification

Table 9 classifies every standardised descriptor. **Category A** (physics-grade inputs): interlayer spacing (Å), scan rate (mV s^{-1}), mass loading (mg cm^{-2}), electrode thickness (μm) and H_2SO_4 concentration (M). Mass loading and electrode thickness qualify despite modest standalone SHAP importance because they appear directly in the verified closure identities, entering the physics term rather than the data term. **Category B** (machine-learning covariates only): layer morphology, composition family and gel electrolyte — informative for prediction, but categorical buckets with no counterpart in any governing relation, and therefore excluded from every physics term. **Category C** (excluded): current density (A g^{-1}), specific surface area ($\text{m}^2 \text{g}^{-1}$), flake size (μm), pore diameter (nm) and synthesis route, on grounds of sparsity, absent variation, or non-convertible units, in addition to all target-derived and metadata columns.

10.4 Proposed formulation

The recommended next-stage model (Fig. 10) predicts $\hat{C}_g$ through a softplus output head, guaranteeing positivity by construction, with auxiliary heads for $\hat{C}_A$ and $\hat{C}_V$. The objective is

$$\mathcal{L} = \lambda_d \mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{phys}} + \lambda_b \mathcal{L}_{\text{bound}} + \lambda_r \|\theta\|^2, \quad (5)$$

with

$$\mathcal{L}_{\text{data}} = \frac{1}{N} \sum_i \text{Huber}(\hat{C}_{g,i}, C_{g,i}) + \sum_{o \in \{A,V\}} \frac{w_o}{|S_o|} \sum_{i \in S_o} \text{Huber}(\hat{C}_{o,i}, C_{o,i}), \quad (6)$$

$$\mathcal{L}_{\text{phys}} = \frac{1}{|S_A|} \sum_{i \in S_A} (\hat{C}_{A,i} - \hat{C}_{g,i} m_{A,i})^2 + \frac{1}{|S_V|} \sum_{i \in S_V} (\hat{C}_{V,i} - \hat{C}_{g,i} m_{A,i}/t_i)^2, \quad (7)$$

$$\mathcal{L}_{\text{bound}} = \frac{1}{N} \sum_i [\max(0, -\hat{C}_{g,i})]^2 + \frac{1}{N} \sum_i d_{\text{hull}}(\mathbf{x}_i)^2, \quad (8)$$

where S_A and S_V index the rows for which the required quantities are actually reported. All physics residuals are masked in this way, so no value is imputed in order to satisfy a constraint. A Huber data term is used rather than squared error because the corpus retains genuine extreme values that a quadratic loss would allow to dominate. Evaluation remains paper-grouped, and no synthetic rows are used.

10.5 What would help most

Table 7 identifies the binding constraint, and the priority list in `data_gap_priorities.csv` ranks the remedies. The two highest-value additions are *potential window* and *discharge time*: at 0% coverage they are the only missing terms of Eq. (1), and recovering them would promote the physics term from a definitional unit closure applicable to a minority of rows to the actual measurement equation applicable to every galvanostatic row. Recording the testing mode would additionally separate galvanostatic from voltammetric measurements, which are currently pooled.

Every split grouped by source paper; no synthetic rows in any evaluation set

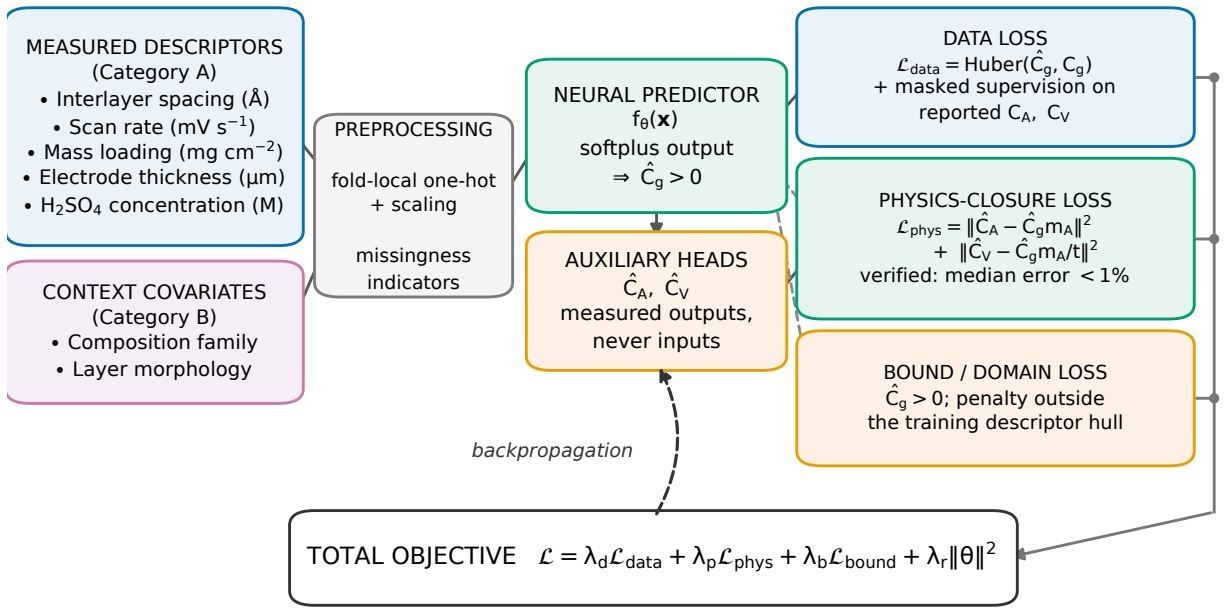


Figure 10: Proposed hybrid physics-constrained multi-task architecture. Category A descriptors enter the predictor; Category B covariates enter the data term only. Areal and volumetric capacitance are auxiliary *outputs*, never inputs, and are tied to the primary output by the closure identities of Eqs. (3)–(4), which were verified against the published values.

11 Discussion

11.1 More rows are not more information

The central finding of this study is a distinction between sample size and independent information. The corpus contains 118 rows but only 40 independent experimental contexts, and 64% of the variance in capacitance lies between those contexts rather than within them. Bayesian-network resampling multiplies rows while holding the number of contexts fixed. Since the generator’s joint distribution is estimated from the training papers alone, every synthetic observation is a recombination of dependencies those papers already exhibit. When the test partition is a *different* publication — with its own protocol, its own mass-normalisation convention and, as Sec. 5.1 shows, frequently its own unique descriptor settings — no amount of resampling can supply the missing information. The augmentation null result and the fidelity success are therefore two aspects of the same fact rather than a contradiction.

This also delimits when the motivating FCDI-style strategy should be expected to work. Where observations are exchangeable draws from one experimental campaign, resampling the joint distribution is a

Table 7: Availability of the quantities required by the standard electrochemical capacitance relations, evaluated over the usable real observations.

Variable	Symbol	n	Coverage (%)	Usable now
h2so4 M	$c0$	111	94	yes
scan rate mV s	nu	88	75	yes
interlayer A	d	88	75	yes
volumetric capacitance F cm3	C_V	65	55	yes
mass loading mg cm2	m_A	55	47	partial - only 47% of rows
electrode thickness um	t	54	46	partial - only 46% of rows
areal capacitance F cm2	C_A	46	39	partial - only 39% of rows
current density A g	j	35	30	partial - only 30% of rows
ssa m2 g	S	35	30	partial - only 30% of rows
electrode area cm2	A	0	0	no - variable is not present in the corpus at all
voltage window V	dV	0	0	no - variable is not present in the corpus at all
current A	I	0	0	no - variable is not present in the corpus at all
discharge time s	dt	0	0	no - variable is not present in the corpus at all
electrode mass g	m	0	0	no - variable is not present in the corpus at all
testing mode	$-$	0	0	no - variable is not present in the corpus at all
cv current integral	$int(IdV)$	0	0	no - variable is not present in the corpus at all

Coverage is the fraction of usable rows for which the quantity can be parsed with an unambiguous unit. The four quantities required by $C_g = I \Delta t / (m \Delta V)$ that are entirely absent from the corpus are the reason a classical PDE-based PINN cannot presently be posed.

Table 8: Empirical verification of the algebraic capacitance closure identities against the values reported in the source publications. These identities are the physics that the corpus can actually enforce.

Closure identity	n rows	Papers	Median err. (%)	< 5%	< 20%	> 50%
$C_A = C_g \cdot m_A$	39	14	0.38	85	90	3
$C_V = C_g \cdot m_A / t$	34	13	0.87	76	85	4

m_A is areal mass loading and t electrode thickness, so m_A/t is the electrode density. Rows failing by more than 50% are dominated by publications whose mass-loading convention differs from the one implied by their reported areal capacitance, which is itself evidence of cross-study protocol heterogeneity.

Table 9: Classification of every standardised descriptor for the next-stage physics-guided model. Category A: physically meaningful, well-covered and stable inputs. Category B: informative machine-learning covariates that are categorical proxies rather than physical parameters. Category C: excluded.

Descriptor	SHAP rank	Miss. (%)	Stability	Category	PINN input?
H ₂ SO ₄ conc. (M)	2	6	stable	A	yes
Scan rate (mV s ⁻¹)	3	25	moderately stable	A	yes
Interlayer spacing (Å)	4	25	moderately stable	A	yes
Electrode thickness (µm)	7	54	stable	A	yes (closure)
Mass loading (mg cm ⁻²)	10	53	stable	A	yes (closure)
Composition family	1	0	stable	B	not directly
Layer morphology	5	19	stable	B	not directly
Gel electrolyte	8	0	unstable	B	not directly
Specific surface area (m ² g ⁻¹)	6	70	moderately stable	C	no
Current density (A g ⁻¹)	9	70	moderately stable	C	no
Pore diameter (nm)	11	97	moderately stable	C	no
Flake size (µm)	12	91	moderately stable	C	no
Synthesis route	-	0	not assessed	C	no

Stability combines the cross-fold SHAP rank spread with the rank change observed between the real-only and Bayesian-augmented models. Full reasoning is in `results_final/pinn/descriptor_classification.csv`.

reasonable regulariser. Where they are clustered by study and evaluation is across studies, it addresses the wrong bottleneck.

11.2 Four purposes that must not be mixed

The analysis maintains four separate roles, and much of the interpretive risk in this problem comes from conflating them. *Predictive assessment* uses paper-grouped cross-validation. *Descriptor interpretation* uses the final real-only model fitted to all observations. *Robustness testing* uses Bayesian augmentation and SHAP-ranking stability. *Physics-model design* uses descriptors supported by all three. In particular, the in-sample fit of the interpretation model is never quoted as performance, and the stability of the SHAP ranking is never taken as evidence that the model predicts well — it does not.

11.3 Attribution, association and causation

SHAP quantifies how a fitted model distributes credit among its inputs. Within this corpus that attribution is stable and reproducible, which makes it a defensible basis for prioritising descriptors. It is not evidence of causation, and Sec. 8 gives a quantitative reason for caution specific to this dataset: several highly ranked descriptors are paper-level constants ($\text{ICC}(1) \approx 1$), so their attribution is statistically entangled with publication identity. The interlayer-spacing trend, which runs opposite to the naive physical expectation, is the clearest symptom. The appropriate conclusion is that these descriptors organise the corpus and the model uses them, not that changing them would change measured capacitance.

11.4 What the physics analysis changes

Rejecting a classical PINN is a substantive result rather than a negative one. It redirects effort from constructing a differential residual the data cannot support towards two things the data do support: enforcing the verified closure identities of Eqs. (3)–(4) as internal-consistency constraints, and extending the extraction schema to recover the potential window and discharge time. The second is the higher-value action. It converts the physics term from a unit-consistency relation applicable to a minority of rows into the measurement equation itself, and it simultaneously addresses the heterogeneity diagnosis, because knowing ΔV and the testing mode removes two known and currently unmodelled sources of between-study scatter.

11.5 Implications for reporting practice

The rows discarded during cleaning are themselves a finding. Capacitance reported only as a range or an inequality, current densities given on incompatible bases, mass loadings without an area, and flake sizes given qualitatively together removed a substantial fraction of otherwise usable records. Standardised reporting of the potential window, testing mode, discharge time and normalisation basis would improve the value of the literature for data-driven analysis more than any modelling refinement available on the present corpus.

12 Limitations

- **Literature-derived, heterogeneous data.** Every row is a structured extraction from a different publication with its own cell geometry, reference electrode, normalisation convention and reporting rate. Two rows with identical descriptors may legitimately differ by a hundred F g^{-1} .
- **Few independent studies.** 40 publications provide 118 rows. Five grouped folds is the practical maximum, and paired significance testing at that scale is underpowered by construction: the smallest attainable two-sided Wilcoxon p -value is 0.0625.
- **Unequal observations per paper.** Cluster sizes range from 1 to 9 rows, so folds differ in effective sample size and a single unusual publication can dominate its fold.
- **Extensive missingness.** Specific surface area, gravimetric current density, flake size and pore diameter are reported by a minority of publications. Requiring complete cases for the Bayesian-network experiment reduced the corpus from 118 to 47 rows, so that experiment describes a sub-population.
- **Protocol inconsistency and unmeasured confounders.** Testing mode, potential window and binder or current-collector details are absent, yet all affect measured capacitance. Some publications

report an interlayer *gap* rather than a (002) *d*-spacing. The closure-identity check found 3 areal and 4 volumetric rows violating an exact definitional relation by more than 50%, which is direct evidence that normalisation conventions differ between publications.

- **Weak cross-paper predictive transferability.** Mean grouped R^2 is 0.030, with negative values on individual folds. The models presented here should not be used to predict the capacitance of an electrode from an unseen laboratory.
- **Descriptor–paper confounding.** Several descriptors are constant within each publication ($\text{ICC}(1) \approx 1$), so their SHAP attribution cannot be separated from publication identity by any analysis of this corpus.
- **Synthetic observations are not experiments.** Bayesian-resampled rows carry no new experimental information and are never treated as independent evidence; they appear in no test partition.
- **SHAP is model attribution, not causality.** All descriptor statements are associations within the compiled corpus.
- **The enforceable physics is definitional.** Equations (3) and (4) are unit-normalisation identities, not transport dynamics. Enforcing them constrains internal consistency; it does not inject electrochemical mechanism, and it applies only to the minority of rows reporting areal or volumetric capacitance.

13 Conclusions

1. **Cross-paper generalisation is weak.** Paper-grouped cross-validation of gravimetric capacitance over 118 observations from 40 publications gives $R^2 = 0.030 \pm 0.119$ and $\text{MAE} = 82 \text{ F g}^{-1}$. A variance decomposition attributes this to cross-study heterogeneity: 64% of the variance in capacitance and a mean of 87% of the variance in the descriptors lies between publications, and several descriptors are paper-level constants.
2. **Bayesian-network augmentation does not robustly improve it.** Across four ratios, three network degrees and three generator seeds, the best condition improved MAE by 6.0 F g^{-1} , less than the 12.3 F g^{-1} produced by re-drawing the synthetic set alone, with a non-monotonic response and improvement in only 3 of 5 folds.
3. **The generator was nevertheless faithful.** No marginal was rejected by a Kolmogorov–Smirnov test, categorical frequencies agreed to a Jensen–Shannon distance of 0.08, dependency structure was reproduced to ≈ 0.06 , and no real row was copied. Distributional fidelity to the training fold does not imply generalisation to an unseen publication.
4. **The SHAP descriptor ranking is stable.** Spearman $\rho = 0.90$ between real-only and augmented models, with 3/3 and 5/5 top- k overlap.
5. **Real-only SHAP is therefore retained for descriptor prioritisation**, with the explicit caveat that highly ranked paper-constant descriptors are confounded with publication identity.
6. **A classical PINN is not currently justified; a hybrid physics-constrained multi-task regression is.** The corpus contains no potential window, discharge time, current or time axis, so no differential residual can be posed. It does contain two exact closure identities, verified here to a median relative error below 1%, which allow areal and volumetric capacitance to act as auxiliary supervised outputs under a hard consistency constraint.
7. **Future effort should target experimental-protocol information rather than synthetic over-sampling.** Recovering the potential window, discharge time and testing mode would convert the physics term into the measurement equation itself and would remove known sources of between-study scatter, addressing the diagnosed bottleneck directly.

Data and code availability

The full pipeline is reproducible from a clean environment with `python run_pipeline.py` followed by `python run_final_stage.py`. All numerical results quoted in this report are read directly from the generated CSV artefacts under `results/` and `results_final/`; package versions and the compatibility patches applied to

DATASYNTHESIZER are recorded in `results/environment.txt`. A fixed random seed of 42 is used throughout.

Nomenclature

C_g	gravimetric capacitance (F/g)
C_A	areal capacitance (F/cm ²)
C_V	volumetric capacitance (F/cm ³)
m_A	areal mass loading (mg/cm ²)
t	electrode thickness (µm)
ρ	electrode density (g/cm ³)
ν	voltammetric scan rate (mV/s)
j	gravimetric current density (A/g)
ΔV	potential window (V)
Δt	discharge time (s)
d	interlayer spacing (Å)
η^2	between-group share of total sum of squares
ICC(1)	one-way intraclass correlation
k	maximum in-degree of the Bayesian network

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